# Tinghan Zhang

## Contact Information

Koopmans Building K543
Department of Econometrics and Operations Research
School of Economics and Management, Tilburg University
Warandelaan 2, 5037 AB, Tilburg, NL

*Tel*: +31-(0)6 2186 3961
*E-mail*: t.zhang_2@tilburguniversity.edu
*Web*: https://tinghan-zhang.github.io/

## Research Interests

Empirical Industrial Organization, Quantitative Marketing, (Applied) Microeconometrics

## Education

**Tilburg University** Tilburg, The Netherlands

*Ph.D. Candidate, Econometrics, School of Economics and Management* *Sep. 2020 - Oct. 2026*

- Supervisors: Prof. dr. Tobias Klein, dr. Christoph Walsh
- Committee: Prof. dr. Bart Bronnenberg (Tilburg University), Prof. dr. Jose Moraga-Gonzalez (VU Amsterdam), dr. Johannes Kasinger (Tilburg University), dr. Ivy Chu Dang (University of Hong Kong)

*Research Master, Economics, School of Economics and Management (Cum Laude)* *Aug. 2018 - Aug. 2020*

**Renmin University of China** Beijing, China

*M.S., Quantitative Economics, Hanqing Advanced Institute of Finance and Economics* *Sep. 2015 - Jul. 2018*

*B.A., Economics and Mathematics, School of Economics* *Sep. 2011 - Jul. 2015*

## Academic Experience

**Teaching Assistant, Tilburg University** 2020 - 2024

*Econometrics 3, Research Master*

- Panel Data, Time Series

*Microeconometrics, Master*

- Treatment Effects, Survival Analysis, Structural Econometrics, Non-/Semi-parametric Estimation

**Research Assistant, Renmin University of China** Jan. 2016 - Jan. 2017

Working for Prof. Guangliang Ye

**Visiting Researcher, Université de Genève** Sep. 2016 - Jan. 2017

Host: Prof. Harold Hau

**Special Auditing Student, Kyoto University** Sep. 2014 - Feb. 2015

Host: Prof. Go Yano

## Working Papers (click to view)

"**A Ranking Representation of Optimal Sequential Search**" (*Job Market Paper, Major Revision at Marketing Science*)

This paper establishes a theoretical equivalence showing that, under the basic assumptions of Independence and Invariance, an optimal sequential search process is observationally equivalent to a partial ranking over all feasible actions throughout the process, thereby introducing a ranking representation of optimal sequential search. This representation yields a simple and unified empirical strategy for implementing sequential search models to extract information from rich data, which was difficult with the traditional policy-based representation. For the classic Weitzman (1979) model, the proposed approach reduces simulation burden while improving estimation accuracy and ease of implementation. The same strategy extends to a broad class of sequential search settings suited for different data structure and search

environments, including partially observed search processes and multi-stage information acquisition, such as sequential search with discovery. Overall, our results enhance the tractability and empirical applicability of sequential search models.

"**Do I Really Want to Buy This? Preference Discovery and Consumer Search**" with Tobias Klein and Christoph Walsh

One of the most invoked assumptions in economics is that consumers know their preferences when making choices. Although theories and experiments in psychology and behavioral economics suggest that this may be unrealistic, there is relatively little evidence from the field on this question. In this paper, we use detailed clickstream data from a large Central Asian online platform to study the extent to which consumers learn about their preferences while searching for a smartphone. To quantify the speed at which this takes place and account for other factors, most notably that consumers obtain additional product information when they inspect product pages, we estimate a rich search model in which consumers learn about their willingness to pay each time they visit the checkout page. Consumers initially underestimate their price sensitivity and update it along the way. Taking this into account shows that consumers are more price-sensitive than a standard search model would predict, and that an intervention that prompts consumers to end their search early can lead to a potential welfare loss.

"**Out of Sight, Out of Cart: A Recall-based Model of Consumer Search**"

We develop a structural sequential search model that incorporates imperfect recall of information acquired during search. Imperfect recall naturally arises in search environments and limits consumers' ability to make optimal purchase decisions. We identify imperfect recall using consumers' revisit actions, which allow them to reacquire decayed information, and estimate the model using rich clickstream data from a large online smartphone marketplace. Our results show that imperfect recall substantially influences both consumers' search process and purchase outcomes. Counterfactual simulations indicate that modest reductions in revisit costs, via mechanisms such as bookmarking, comparison tools, or retargeting prompts, can meaningfully reduce consumers' suboptimal purchases. These findings highlight the role of imperfect recall in consumer search and provide guidance for interventions that improve consumers' decision quality.

## Selected Work in Progress

"Better Price, Better Quality? Resolve the Endogeneity in Search Decisions"

"Search Cost, Risk, and Financial Institutions Merger," with Chenxi Wang

## Conferences and Seminars

2026: Barcelona Quant Marketing Workshop (invited discussant); IAAE 2026; 15th Consumer Search and Switching Costs Workshop; APIOC 2026 (scheduled).

2025: EMAC Doctoral Colloquium; INFORMS Marketing Science Conference 2025; 14th Consumer Search and Switching Costs Workshop; EWMES 2025; APIOC 2025*; AMA Summer Conference 2025*; EEA 2025*.

2021-24: Tilburg Structural Econometric Group; Tilburg Graduate Students Society Seminar; Workshop on Digital Markets 2024.

** Paper accepted but not able to present due to time conflicts.*

## Honors and Awards

EMAC Doctoral Colloquium Fellow 2025
Tilburg University: Koopmans Scholarships, 2018 - 2020
Renmin University of China: Excellent Student Scholarships, 2015-2017
Renmin University of China: Excellent Graduate, 2018, 2015
Kyoto University: JASSO Scholarships, 2014

## Other Activities

**China Banking Regulatory Commission** Beijing, China
*Intern in Bureau of Policy Research* *Jun 2015 - Aug 2015*

**Tilburg Structural Econometrics Group** Tilburg, Netherlands
*Co-organizer of the Lunch Seminar Series* *Sep 2022 - Jun 2024*

## Technical Skills

**Languages**: Chinese (Native), English (Proficient), Japanese (Advanced), Dutch (Beginner)
**Research Software**: Matlab, Julia, Stata, R
**Application Software**: LaTeX, Jupiter Notebook, Git

## Contact Information of References

**Tobias J. Klein**
Department of Econometrics and OR
Tilburg University
T.J.Klein@tilburguniversity.edu

**Bart J. Bronnenberg**
Department of Marketing
Tilburg University
Bart.Bronnenberg@tilburguniversity.edu

**Christoph B. T. Walsh**
Department of Econometrics and OR
Tilburg University
C.B.T.Walsh@tilburguniversity.edu